MIL-M-45423C (MU)
6 JUNE 1973
SUPERSEDING
MIL-M-45423B
15 JULY 1970

MILITARY SPECIFICATION

MINE, APERS, M18A1 (WITH ACCESSORIES) LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of mine designated as Mine, Apers, M18A1, (formerly T48E3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Method of
MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables For Inspection by Attributes.

FSC: 1345

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8800900 - Box, Wirebound, Packing, Ammunition for Mine, APERS, M18A1 or Mine, APERS, Practice XM-68 with Accessories
8800901 - Carton, Packing, Ammunition for Mine, APERS, M18A1 or Mine, APERS, Practice, XM-68 with Accessories
8835166 - Mine, APERS, M18A1, with Accessories
8837104 - Mine APERS, M18A1 Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Mines. - The mines with accessories shall comply with all requirements specified on Drawing (dwg.) 8835166, and with all requirements specified in applicable specifications.

3.3 Functioning. - The loaded mine assembly with accessories, following temperature-conditioning, shall function satisfactorily as evidenced by completely destroying itself. The test shall be performed at an approved government facility.

3.4 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible

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for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot Formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements.

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The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Parts of one lot interfix number from one manufacturer.
b. Composition C-4 from not more than two lots.
c. Firing devices of one lot interfix number from one manufacturer.
d. Electrical test sets of one lot interfix number from one manufacturer.
e. Electrical blasting cap assemblies of one lot interfix number from one manufacturer.

4.2.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Assembly, Mine, APERS, M18A1, prior to sealing (see dwg. 8837104 covering a detail of dwg. 8835166).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None Defined		
Major:	AQL 0.40 percent		
101.	Gap between explosive and inside surface of sidewall, maximum	Gage	01001
Minor:	AQL 1.5 percent		
201.	Bead of Cement applied less than 360°	Visual	01002
202.	Edges chipped or broken	Visual	01003
203.	Locking lug missing or damaged	Visual	01004
204.	Evidence of poor workmanship (see 3.4)	Visual	01005

4.2.2.2 Assembly, Mine, APERS, M18A1, after sealing (see dwg. 8837104 covering a detail of dwg. 8835166).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 1.0 percent		
101.	Detonator well depth, minimum	Gage	02001
102.	Detonator well diameter, minimum	Gage	02002
103.	Shipping plug-priming adapter missing, damaged or improperly assembled	Visual/ Manual	02003
104.	Leg assembly twisted, bent or otherwise damaged	Visual	02004
105.	Crimp missing or within inside wall surface	Visual	02005
Minor:	AQL 1.5 percent		
201.	Crack or split	Visual	02006
202.	Marking misleading or unidentifiable	Visual	02007
203.	Evidence of poor workmanship (see 3.4)	Visual	02008

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4.2.2.3 Mine, APERS, M18A1 with accessories (see dwg. 8835166).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Mine or accessory missing	Visual	03001
102.	Mine or accessory damaged	Visual	03002
103.	Mine or accessory incorrectly packed	Visual	03003
104.	Tag damaged or missing from bandoleer containing test set	Visual	03004
Minor:	AQL 1.0 percent		
201.	Marking misleading or unidentifiable	Visual	03005
202.	Evidence of poor workmanship (see 3.4)	Visual	03006

4.2.2.4 Carton, prior to sealing (see dwg. 8800901)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Number of field packs incorrect	Visual	08001
102.	Field pack containing M40 test set missing or incorrectly positioned in carton	Visual	08002
103.	Packing component missing	Visual	08003
Minor:	None defined		

4.2.2.5 Carton, sealed (see dwg. 8800901).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.40 percent		
101.	Contents move when carton is shaken (insufficient filler material)	Visual/ Manual	04001
102.	Carton damaged so that contents are exposed or liable to be exposed	Visual/ Manual	04002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	04003

4.2.2.6 Bag, sealed barrier (see dwg. 8800901)

Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.40 percent		
101.	Waterproofness destroyed by poor seal or damage (separation of heat seal, rip, or puncture)	Visual/ Manual	05001
Minor:	AQL 1.5 percent		
201.	Heat seal incorrectly located	Visual	05002
202.	Air not removed from bag	Visual/ Manual	05003
203.	Marking misleading or unidentifiable	Visual	05004

4.2.2.7 Box, packing, sealed wooden (see dwg. 8800900)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Wire loose or improperly stapled	Visual/ Manual	06011
102.	Board broken	Visual	06003

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Minor:	AQL 1.5 percent		
201.	Wire loop incorrectly closed	Visual	06012
202.	Car seal missing, unsealed or improperly positioned	Visual	06009
203.	Marking misleading or unidentifiable	Visual	06010

4.2.3 Testing

4.2.3.1 Functioning (see 3.3) - Major defect - Code No. 07001. - The sampling plan for this test shall be in accordance with MIL-STD-105, Code letter G, double sampling plan and an AQL of 1.5 percent. The test shall be performed as specified in 4.3.1 using equipment in accordance with 4.2.4. Any assembly which fails to comply with the specified requirement shall be classed defective.

4.2.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 (see 6.3).

4.3 Test methods and procedures.

4.3.1 Functioning. - This test shall be performed at an approved Government facility. The loaded mine and the accessories contained in its bandoleer shall be conditioned at minus 40, plus or minus 2 degrees, Fahrenheit for 25 plus or minus 5 hours. It shall then be functioned with the conditioned accessories within 3 minutes after removal from the temperature-conditioning chamber. Functioning shall be in accordance with the instruction sheet sewn on the bandoleer. Observation shall be made to determine compliance with the requirement specified in 3.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. - Packaging for Mine, APERS, M18A1, with accessories, dwg. 8835166, shall be in accordance with dwg. 8800901. Accessories which shall be included are:

a. Firing Device, Electrical, M57
b. Test Set, Electrical Circuit, M40
c. Cap, Blasting, Electrical, M4
d. Bandoleer, M7

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5.2 Packing.

5.2.1 Level A. - Packing for Mine, APERS, M18A1, with accessories, shall be in accordance with dwg. 8800900.

5.3 Marking. - Unit and exterior containers shall be marked in accordance with instructions contained in drawings 8800900 and 8800901.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.2 Inspection Code Numbers. - The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs. - Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

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c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1345-A-017

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